# T2K First-Year $\nu_\mu$-Disappearance Analysis

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**Abstract**

This note summarizes work by the Rutherford-Oxford groups synthesizing inputs from T2KSK, nd280, beam and neutrino generator (NEUT, GENIE) working groups and performing a $\nu_\mu$-disappearance analysis on the first-year T2K data.

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## 1. Introduction

## 2. Methodology

### *2.1. SuperK reconstructed energy spectrum prediction*

#### *2.1.1. Outline*

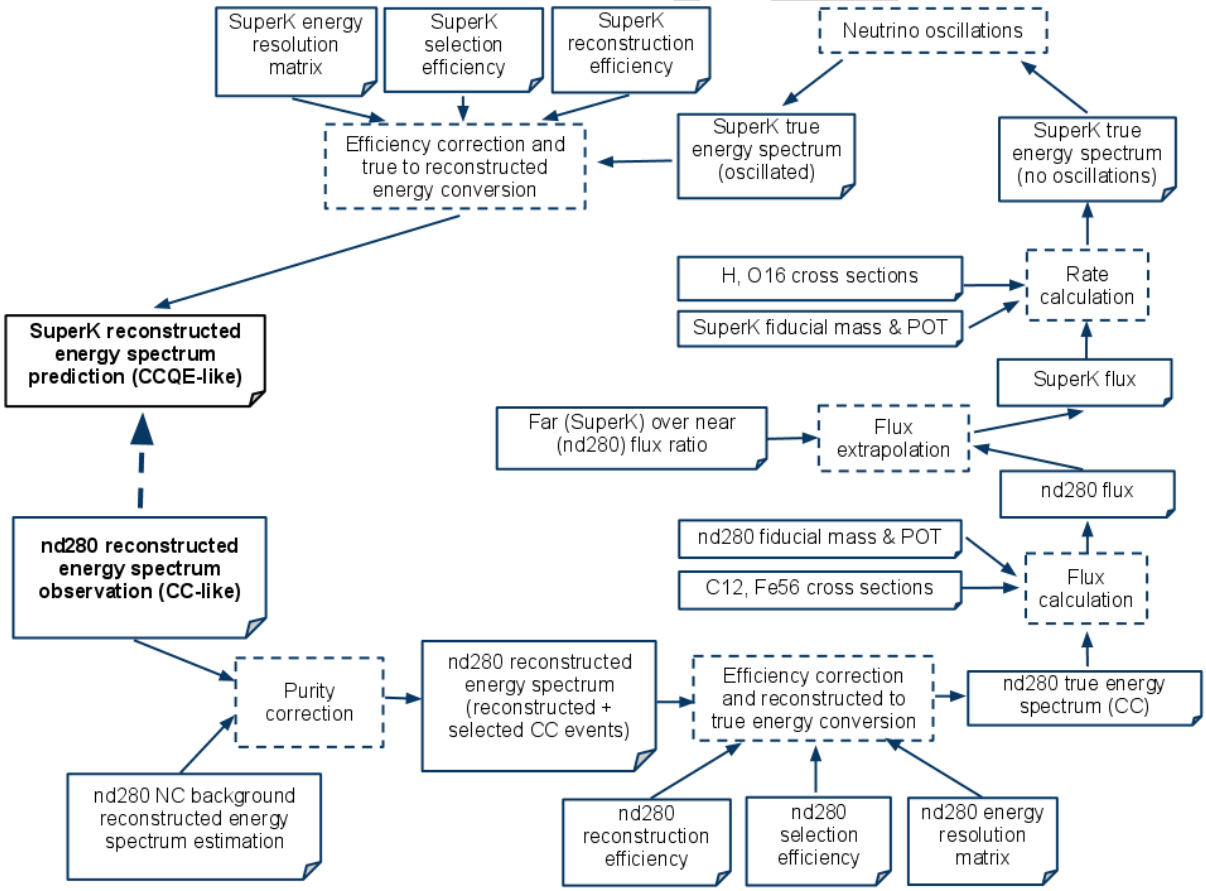



Figure 1: Outline of the procedure for predicting the SuperK reconstructed energy spectrum.

#### *2.1.2. Detailed derivation of SuperK spectrum prediction*

The expected reconstructed energy ($E_r$) spectrum ($\frac{dN^{exp}_{SK}}{dE_r}$) at Super-K, for selected neutrino events is the sum of the corresponding spectra for all (mutually exclusive) samples $s$ included in the analysis ($s$ = FCFV 1-ring $\mu$-like, FCFV multi-ring $\mu$-like, ...) and all event types $m$ ($m = \nu_\mu$ CCQE, $\nu_\mu$ non-CCQE, NC) contributing to each sample:

$$\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \sum_s \frac{dN^{exp}_{SK;s}(E_r)}{dE_r} = \sum_{s;m} \frac{dN^{exp}_{SK;s;m}(E_r)}{dE_r} \quad (1)$$

Let $\epsilon^{reco}_{SK;m}(E_t)$ be the Super-K efficiency for reconstructing a neutrino event of type $m$ and true energy $E_t$, and $\epsilon^{sel}_{SK;s;m}(E_r)$ be the Super-K efficiency for selecting an event of type $m$ and reconstructed energy $E_r$ in sample $s$. If $R_{SK;s;m}(E_t, E_r)$ is the Super-K energy resolution matrix for events of type $m$ in sample $s$, the expected reconstructed energy spectrum can be expressed in terms of the total true neutrino energy spectrum ($\frac{dN^{tot}_{SK;s;m}(E_t)}{dE_t}$):

$$\frac{dN_{SK}^{exp}(E_r)}{dE_r} = \sum_{s;m} \int dE_t \frac{dN_{SK;s;m}^{tot}(E_t)}{dE_t} R_{SK;s;m}(E_t, E_r) \epsilon_{SK;s;m}^{sel}(E_r) \epsilon_{SK;m}^{reco}(E_t) \tag{2}$$

The above expression can be rewritten to explicitly include the survival probability $P_m^{surv}(E_t)$ for neutrinos of energy $E_t$, and the total unoscillated true neutrino energy spectrum $\frac{dN_{SK;s;m}^{tot;unosc}}{dE_t}$:

$$\frac{dN_{SK}^{exp}(E_r)}{dE_r} = \sum_{s;m} \int dE_t \frac{dN_{SK;s;m}^{tot;unosc}(E_t)}{dE_t} P_m^{surv}(E_t) R_{SK;s;m}(E_t, E_r) \epsilon_{SK;s;m}^{sel}(E_r) \epsilon_{SK;m}^{reco}(E_t) \tag{3}$$

The total unoscillated true neutrino energy spectrum is related to the estimated neutrino flux at Super-K ($\frac{d\Phi_{SK}(E_t)}{dE_t}$) in the following way:

$$\begin{aligned} \frac{dN_{SK}^{exp}(E_r)}{dE_r} = N_{SK}^{POT} M_{SK}^{fid} \frac{N_A}{A_{H20}} \sum_{s;m} \int dE_t \Bigg( & \frac{d\Phi_{SK}(E_t)}{dE_t} \times \\ & \sigma_{H20}^{tot}(E_t) f_m(E_t) P_m^{surv}(E_t) R_{SK;s;m}(E_t, E_r) \epsilon_{SK;s;m}^{sel}(E_r) \epsilon_{SK;m}^{reco}(E_t) \Bigg) \end{aligned} \tag{4}$$

where $N_{SK}^{POT}$ is the proton-on-target exposure for the analyzed Super-K data sample, $M_{SK}^{fid}$ is the Super-K fiducial volume mass, $\sigma_{H20}^{tot}(E_t)$ is the total neutrino interaction cross section in water at energy $E_t$, $f_m(E_t$ is the fraction of events of type $m$, $N_A$ is Avogadro's number and $A_{H20}$ is the water mass number.

The absolute neutrino flux $\frac{d\Phi_{SK}(E_t)}{dE_t}$ in Eq. 4, for which significant uncertainties exist, can be constrained if it is expressed in terms of the observed reconstructed neutrino energy spectrum from an ND280 inclusive charged-current sample. From the definition of the far-over-near ratio

$$\frac{d\Phi_{SK}(E_t)}{dE_t} = R^{F/N}(E_t) \frac{d\Phi_{nd280}(E_t)}{dE_t} \tag{5}$$

where $\frac{d\Phi_{nd280}(E_t)}{dE_t}$ is the neutrino flux at ND280. This latter quantity can be related to the total CC true neutrino energy spectrum $\frac{dN_{nd280;CC}^{tot}(E_t)}{dE_t}$ as follows:

$$\frac{d\Phi_{SK}(E_t)}{dE_t} = \frac{R^{F/N}(E_t) A_{nd280}}{N_{nd280}^{POT} M_{nd280}^{fid} N_A \sigma_{nd280}^{CC}(E_t)} \frac{dN_{nd280;CC}^{tot}(E_t)}{dE_t} \tag{6}$$

where $N_{nd280}^{POT}$ is the proton-on-target exposure for the analyzed ND280 data sample, $M_{nd280}^{fid}$ is the fiducial mass for the ND280 'target' region, $\sigma_{nd280}^{CC}(E_t)$ is the total charged-current cross section for the ND280 'target' region mixture at energy $E_t$ and $A_{nd280}$ is the mass number for the ND280 'target' mixture.

The total CC true neutrino energy spectrum can be expressed in terms of the observed reconstructed neutrino energy spectrum (which has the NC background subtracted):

$$\begin{aligned} \frac{d\Phi_{SK}(E_t)}{dE_t} = \frac{R^{F/N}(E_t) A_{nd280}}{N_{nd280}^{POT} M_{nd280}^{fid} \sigma_{nd280}^{CC}(E_t)} \int dE_{r\prime} \Bigg( & \\ \left( \frac{dN_{nd280;CC-like}^{obs}(E_{r\prime})}{dE_{r\prime}} - \frac{dN_{nd280;NC-bkg}^{exp}(E_{r\prime})}{dE_{r\prime}} \right) & R_{nd280;CC}^{T}(E_t, E_{r\prime}) \frac{1}{\epsilon_{nd280;CC}^{reco}(E_t) \epsilon_{nd280;CC}^{sel}(E_{r\prime})} \Bigg) \end{aligned} \tag{7}$$

where $\epsilon_{nd280;CC}^{reco}(E_t)$ is the ND280 efficiency for reconstructing a charged-current neutrino event with true energy $E_t$, $\epsilon_{nd280;CC}^{sel}(E_{r\prime})$ is the ND280 selection efficiency for a charged-current event reconstructed with energy $E_{r\prime}$, and $R_{nd280;CC}^{T}(E_t, E_{r\prime})$ is the transpose of the ND280 energy resolution matrix for charged-current events.

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Combining Eq. 4 and 7 we obtain

$$\begin{aligned}\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \frac{N^{POT}_{SK} M^{fid}_{SK} A_{nd280}}{N^{POT}_{nd280} M^{fid}_{nd280} A_{H20}} \sum_{s;m} \int dE_t \int dE_{r\prime} \Bigg( R^{F/N}(E_t) \frac{\sigma^{tot}_{H20}(E_t) f_m(E_t)}{\sigma^{CC}_{nd280}(E_t)} P^{surv}_m(E_t) \times \\ \left( \frac{dN^{obs}_{nd280;CC-like}(E_{r\prime})}{dE_{r\prime}} - \frac{dN^{exp}_{nd280;NC-bkg}(E_{r\prime})}{dE_{r\prime}} \right) \times \\ R_{SK;s;m}(E_t, E_r) R^T_{nd280;CC}(E_t, E_{r\prime}) \frac{\epsilon^{sel}_{SK;s;m}(E_r) \epsilon^{reco}_{SK;m}(E_t)}{\epsilon^{reco}_{nd280;CC}(E_t) \epsilon^{sel}_{nd280;CC}(E_{r\prime})} \Bigg)\end{aligned} \tag{8}$$

In the special case where no spectral information is available from nd280 for the 1st year analysis, Eq. 8 becomes

$$\begin{aligned}\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \frac{N^{POT}_{SK} M^{fid}_{SK} A_{nd280}}{N^{POT}_{nd280} M^{fid}_{nd280} A_{H20}} \sum_{s;m} \int dE_t \Bigg( R^{F/N}(E_t) \frac{\sigma^{tot}_{H20}(E_t) f_m(E_t)}{\sigma^{CC}_{nd280}(E_t)} P^{surv}_m(E_t) \times \\ \left( N^{obs}_{nd280;CC-like} - N^{exp}_{nd280;NC-bkg} \right) R_{SK;s;m}(E_t, E_r) \frac{\epsilon^{sel}_{SK;s;m}(E_r) \epsilon^{reco}_{SK;m}(E_t)}{\epsilon_{nd280;CC}(E_t)} \Bigg)\end{aligned} \tag{9}$$

*2.1.3. Discussion*

*2.1.4. Summary of required inputs*

*From SuperK:*

- $\epsilon^{reco}_{SK;m}(E_t)$: efficiency for reconstructing a neutrino event of type $m$ and true energy $E_t$.
- $\epsilon^{sel}_{SK;s;m}(E_r)$: efficiency for selecting an event of type $m$ and reconstructed energy $E_r$ in sample $s$.
- $R_{SK;s;m}(E_t, E_r)$: energy resolution matrix for events of type $m$ in sample $s$.
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*From nd280:*

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-
-

*From JNUBEAM:*

- $R^{F/N}(E_t)$: Far over (off axis) near flux ratio in terms of true energy $E_t$.
-
-

*From GENIE/NEUT:*

- $\sigma^{tot}_{H20}(E_t)$: total muon neutrino interaction cross section in water at energy $E_t$.
- $f_m(E_t)$: fraction of events of type $m$ for muon neutrino interactions in water.
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*2.2. Oscillation fitting*

*2.2.1. Fitting method*

Measurements of the oscillation parameters $sin^2 2\theta_{23}$, $\Delta m^2_{23}$ are obtained by comparing the observed SuperK reconstructed energy spectrum with the SuperK spectrum that is predicted using the near detector measurements; this prediction is made using the method described in section 2.1. Let $N$ be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the SuperK reconstructed energy spectrum prediction (e.g parameters that control the beam and neutrino interaction simulations and the detector response). Best-fit values are obtained by minimizing:

$$\chi^2(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = 2 \cdot \sum_{i=0}^{N-1} \left( n_i^{obs} \cdot ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp} - n_i^{obs}) \right) + (\mathbf{a} - \mathbf{a_0})^T \cdot \mathbf{C}^{-1} \cdot (\mathbf{a} - \mathbf{a_0}) \quad (10)$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of considered systematics parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, $n_i^{obs}$ is the observed number of events in the $i^{th}$ bin, and $n_i^{exp} = n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ is the corresponding expected number of events.

Due to the limited statistics in SuperK and the limited amount of information available to this analysis from the near detectors, we do not fit for systematics. Instead, we evaluate oscillation systematics as described in section 2.3.

*2.2.2. Comment on the differences between the MINOS and K2K fitting methods*

In discussions of this work, questions were raised regarding the differences between our fitting method and the binned maximum likelihood method employed in the alternative T2K disappearance analysis by J.Kameda et al. The fitting method used in this work is the method used in the MINOS disappearance analysis [1], while the method used by J.Kameda et al. is the method used in the K2K disappearance analysis [2]. In this section we demonstrate that the two methods are equivalent. More detailed information can be found in [3].

Let $\mathbf{n^{obs}}$ be a set of $N$ measurements (the number of events in each of the $N$ bins). The probability density function (p.d.f.) for the total number of events $n_{tot}^{obs} = \sum_{i=0}^{N-1} n_i^{obs}$ observed in SuperK has a Poisson distribution, and the mean of that distribution is equal to the predicted total number of events $n_{tot}^{exp} = n_{tot}^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = \sum_{i=0}^{N-1} n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$. The p.d.f. is:

$$f_{poisson} = f_{poisson}(\mathbf{n^{obs}}; sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = \frac{[n_{tot}^{exp}]^{n_{tot}^{obs}} \, e^{-n_{tot}^{exp}}}{n_{tot}^{obs}!} \quad (11)$$

The reconstructed energies of the observed events are distributed over $N$ bins. Since there are $N$ possible outcomes for the process of placing an event in a bin, the p.d.f. of the distribution of those outcomes is multinomial:

$$f_{multinomial} = f_{multinomial}(\mathbf{n^{obs}}; sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = n_{tot}^{obs}! \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} P_i^{n_i^{obs}} \quad (12)$$

where $P_i = P_i(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ is the probability that an event is placed in the $i^{th}$ bin. This probability can be calculated as the ratio of the number of events predicted for that bin to the predicted total number of events:

$$P_i = \frac{n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})}{n_{tot}^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})} \quad (13)$$

Therefore, $f_{multinomial}$ can be written as:

$$f_{multinomial} = n_{tot}^{obs}! \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} \left[ \frac{n_i^{exp}}{n_{tot}^{exp}} \right]^{n_i^{obs}} \quad (14)$$

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The joint p.d.f. $f_{joint} = f(\mathbf{n^{obs}}; sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ for the outcome of the reconstructed energy distribution is the product of the Poisson and multinomial distributions shown in Eq. 11 and Eq. 14 respectively:

$$f_{joint} = e^{-n_{tot}^{exp}} \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} \left[n_i^{exp}\right]^{n_i^{obs}} \tag{15}$$

The p.d.f. in Eq. 15 is identical ?? to the likelihood function used by K2K in [2]: $L_{NORM}$ corresponds to the Poisson term and $L_{SHAPE}$ to the multinomial term. K2K obtains the best-fit values of the oscillation parameters by taking the natural logarithm of equation **??**, and maximising it with respect to those parameters.

However the values of the best-fit oscillation parameters are unchanged if the p.d.f. in Eq. 15 is multiplied or divided by a factor that does not depend on the oscillation parameters. In the present analysis, this p.d.f. is divided by

$$f_0 = f_0(\mathbf{n^{obs}}) = \frac{\left[n_{tot}^{obs}\right]^{n_{tot}^{obs}} e^{-n_{tot}^{obs}}}{n_{tot}^{obs}!} n_{tot}^{obs}! \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} \left[\frac{n_i^{obs}}{n_{tot}^{obs}}\right]^{n_i^{obs}} \tag{16}$$

which simplifies to

$$f_0 = e^{-n_{tot}^{obs}} \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} \left[n_i^{obs}\right]^{n_i^{obs}} \tag{17}$$

Dividing Eq. 15 by Eq. 17 gives the quantity $\lambda$ (likelihood ratio):

$$\lambda = e^{n_{tot}^{obs} - n_{tot}^{exp}} \prod_{i=0}^{N-1} \left(\frac{n_i^{exp}}{n_i^{obs}}\right)^{n_i^{obs}} \tag{18}$$

In the large-sample limit, the quantity

$$-2\ln\lambda = 2 \sum_{i=0}^{N-1} \left[ n_i^{obs} \ln\left(\frac{n_i^{obs}}{n_i^{exp}}\right) + n_i^{exp} - n_i^{obs} \right] \tag{19}$$

has a $\chi^2$ distribution. As it can be seen, the first term of Eq. 10 is identical to -2 ln$\lambda$ shown in Eq. 19.

### *2.3. Evaluating oscillation systematics*

#### *2.3.1. Evaluation method*

#### *2.3.2. Handling correlated systematics*

#### *2.3.3. Cholesky decomposition method*

The LU decomposition is a well-known factorization of a matrix $\mathbf{C}$ as a product of a lower triangular matrix $\mathbf{L}$ and an upper triangular matrix $\mathbf{U}$. The LU decomposition has several applications in scientific computing [4]. Let $\mathbf{C}$ be a square, symmetric, positive-definite[1] matrix. Under the stated conditions, there exists special factorization of $\mathbf{C}$, the Cholesky decomposition, which is significantly *more efficient* than the generic LU decomposition. Rather than factorizing $\mathbf{C}$ as product of two independent matrices $\mathbf{L}$ and $\mathbf{U}$, the Cholesky method factorizes $\mathbf{C}$ as product of a lower triangular matrix $\mathbf{L}$ and its transpose matrix:

$$\mathbf{C} = \mathbf{L} \cdot \mathbf{L}^T \tag{20}$$

The Cholesky decomposition algorithm can be found in [4]. Code is also available in ROOT[2].

The Cholesky decomposition is commonly used for Monte Carlo simulation of systems with correlated variables. Let $S_N$ be such a system whose dimensionality is $N$ and let $\mathbf{C}$ be the $N \times N$ matrix which

[1] $bfv \cdot \mathbf{C} \cdot \mathbf{v} > 0$ for every vector $\mathbf{v}$.
[2] See TDecompChol class.

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describes the covariance of the $N$ simulated variabled (covariance matrix). The covariance matrix $\mathbf{C}$ is decomposed to obtain the lower triangular matrix $\mathbf{L}$. Let $\mathbf{u}$ be a $N$-dimensional vector of uncorrelated variables; each vector element $u_i$ can be randomly generated from a gaussian distribution with a mean of 0 and a standard deviation of 1. The vector

$$\mathbf{x} = \mathbf{L} \cdot \mathbf{u} \tag{21}$$

has the covariance properties of $S_N$.

## 3. Analysis

## 4. Summary

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